

# Sushil Panthi

Full Stack Developer (MERN & APIs) | Agentic AIAgentic AI and Backend Engineer | Power BI and Data Analytics Developer | Open to Relocation

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## 📄 PROFILE

Results-driven Software Developer and BCA graduate (CGPA: 8.6) with hands-on experience across 4 internships and 8+ scalable projects. I specialize in building production-ready MERN applications and advanced Agentic AI workflows using LangChain and LangGraph. With 4 international research publications—including a Best Paper Award—and expertise in deploying FastAPI services on AWS, I bridge the gap between academic excellence and real-world engineering. Committed to architecting clean, user-centric solutions that leverage data analytics and AI to drive measurable impact.

## 📁 PROFESSIONAL EXPERIENCE

- Agentic AI & Data Science (GenAI) Intern**, Innomatics Research Labs

11/2025 – 05/2026  
Remote

  - Developed a stateful customer support assistant using LangGraph to manage complex decision-making nodes, conditional routing, and managed query lifecycles.
  - Engineered a localized RAG pipeline utilizing ChromaDB for vector storage and recursive character splitting to ensure document-grounded, factually accurate responses.
  - Implemented Human-in-the-Loop (HITL) safety gates to identify knowledge-gap or low-confidence queries and route them to human agents with explicit reasoning.
  - Optimized multi-model inference by integrating Groq for low-latency generation with a Gemini fallback strategy and full execution transparency via LangSmith.
- Tech Intern**, Anishk Sustainable Development Foundation (ASDF)

04/2024 – 04/2025  
Remote

  - Collaborated with a team to develop the "Arka Journal" website, focusing on research and publications related to rural and tribal community initiatives.
  - Utilized HTML, CSS, JavaScript, and PHP to build a responsive, user-friendly website for showcasing research papers and publications.
  - Worked closely with developers, designers, and researchers to integrate research content seamlessly into the website.
  - Enhanced the accessibility and visibility of community-driven research by creating a digital platform for sharing knowledge and innovations.
- Full Stack Development Intern**, Innomatics Research Labs

01/2025 – 03/2025  
Remote

  - Developed dynamic, component-based UIs using React.js and implemented secure server-side logic with Node.js and Express.js for real-world web applications.
  - Designed and optimized MongoDB schemas and queries for efficient data modeling, directly improving application performance and reliability.
  - Delivered LumbiniCare Connect as the capstone project — a full-stack hospital management system with JWT-based role access, RESTful APIs, and a responsive Material-UI frontend.
  - Applied full-stack best practices end-to-end: from UI component architecture and API design to database integration and deployment.
- Full Stack Development Intern**, Pantech Prolabs India Pvt Ltd

07/2024 – 01/2025  
Remote

  - Gained hands-on experience building full-stack web applications using the MERN stack — developing React.js interfaces, Node.js/Express.js APIs, and MongoDB databases from the ground up.
  - Debugged, tested, and optimized code across frontend and backend layers, developing a strong foundation in scalable application development.
  - Applied software development best practices including component architecture, RESTful API design, and Git version control in a structured, real-world environment.

## 📁 PROJECTS

- SkillNexus - Learn from Humans (College Final Year Project)**,  
Using React, Vite, React Router (HashRouter), Axios, Node.js, Express, MongoDB, Mongoose, JWT, CSS utility classes
- Built a peer-to-peer learning marketplace where people learn directly from humans (not prerecorded videos), with mobile- first UI, width-safe layouts.
  - Engineered a unified friendship + notification lifecycle with request/accept/reject/cancel endpoints, auto-delete pending notifications on cancel, single-item status that persists across refresh, relation-based syncing, duplicate suppression, and event-driven/visibility refresh.
  - Implemented secure, reliable platform foundations using JWT auth, protected admin areas with token validation, Mongoose data models, scrollable one-line filters for clean discovery, consistent purple brand styling, accessible contrast, and larger tap targets.
  - Delivered end-to-end performance and quality with removed noisy SSE 404s, optimized axios base/proxy, added toasts and disabled buttons during network calls, ran lint/QA across routes, and verified production behavior post-deploy.

**AlgoCare - Reducing the Wait, Prioritizing the Life (Hackathon Project),**

Using React, Vite, Node.js (ESM), Express, Socket.io, MongoDB, Tailwind/MUI

- Developed an intelligent hospital resource and queue management SaaS for Track 3 of HackGenX 2026, delivering tailored dashboards for Patients, Doctors, and Administrators.
- Engineered a self-correcting queue engine using a Moving Average Algorithm that calculates high-precision patient ETAs based on the actual completion speed of the last five consultations.
- Implemented zero-latency real-time synchronization using Socket.io, enabling instant cross-dashboard updates for token status, bed availability, and resource levels without page refreshes.
- Built a secure, modular foundation using a Controller-Service-Repository pattern with JWT-based role-based access control (RBAC) and automated history linking via email-based patient identification.

Source code [↗](#)

**Emotion-Based Mental Health Tracker,**

Using React, TypeScript, Vite, Material-UI, Recharts, JWT, localStorage (simulated DB), Tailwind CSS

- Emotion-first risk assessment with automatic webcam emotion analysis on login, 8-question survey, instant personalized risk score and recommendations.
- Closed-loop improvement workflow with camera-monitored guided activities, 5-question emotion retest, quantified pre/ post improvement and visual comparisons.
- Iterative personalized recommendations via dynamic re-assessment loop delivering targeted interventions and next-step guidance to reduce user risk.
- Secure analytics-backed platform with role-based authentication, protected admin dashboard, persistent localStorage data and exportable real-time dashboards.

Source code [↗](#)

**LumbiniCare Connect: Lumbini Nepal Hospital,** Using MERN Stack (MongoDB, Express.js, React.js, Node.js) Source Code

- Built a full-stack hospital management system enabling efficient management of appointments, prescriptions, and departments.
- Implemented secure authentication and role-based access control (RBAC) using JWT for data privacy and restricted access.
- Designed scalable RESTful APIs for CRUD operations with robust error handling and MongoDB integration.
- Enhanced user experience with a responsive React.js frontend and Material-UI for a mobile-friendly interface.

Source code [↗](#)

 **SKILLS**

**Full Stack Development** — MERN Stack (MongoDB, Express.js, React.js, Node.js), REST APIs, JWT, Socket.io, MVC / Controller-Service-Repository

**Web Technologies & UI** — HTML5, CSS3, Tailwind CSS, Material-UI, Vite, Responsive Design, Accessibility (a11y)

**Data Analytics & Power BI:** — Power BI, Pandas, NumPy, Matplotlib, Data Visualization, Microsoft Excel

**AI / GenAI** — Agentic AI, LangChain, CrewAI, RAG, Prompt Engineering, Generative AI, LLM Integration (GPT-4, Claude API)

**Programming Languages** — JavaScript (ES6+), TypeScript, Python, Java, SQL, C, C++

**Databases** — MongoDB, MySQL, Mongoose ODM, RESTful API Design, CRUD, Indexing

**Tools & Platforms** — Git, GitHub, Postman, VS Code, Agile/Scrum, Microsoft Office Suite

**Soft Skills** — Leadership, Cross-functional Collaboration, Problem Solving, Rapid Prototyping, Public Speaking, Leadership, Communication

 **EDUCATION**

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|-------------------------------------------------------------------------------|----------------------------|
| <b>Bachelor of Computer Applications (BCA) (CGPA: 8.6),</b> Sandip University | 07/2023 – 06/2026          |
|                                                                               | Nashik, Maharashtra, India |

 **RESEARCH PAPER**

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| <b>(ICETDA 2025), 3rd International Conference on Emerging Trends of Design &amp; Arts at Poornima University</b><br>Jaipur, Rajasthan                                                                 | 01/08/2025 |
| <b>(FINCON 2025), International Financial Security and Management Conference-2025 at National Forensic Sciences University</b><br>Gandhinagar, Gujarat                                                 | 01/07/2025 |
| <b>International Conference on Advances in Computational Intelligence and Applications-2025 at Insitute of Information Technology &amp; Management(IITM), GGS Indraprastha University</b><br>New Delhi | 01/10/2025 |
| <b>IEEE Conferences- ARRIA 2024 at Manipal Institute of Technology</b><br>Manipal, Karnataka                                                                                                           | 01/11/2024 |